

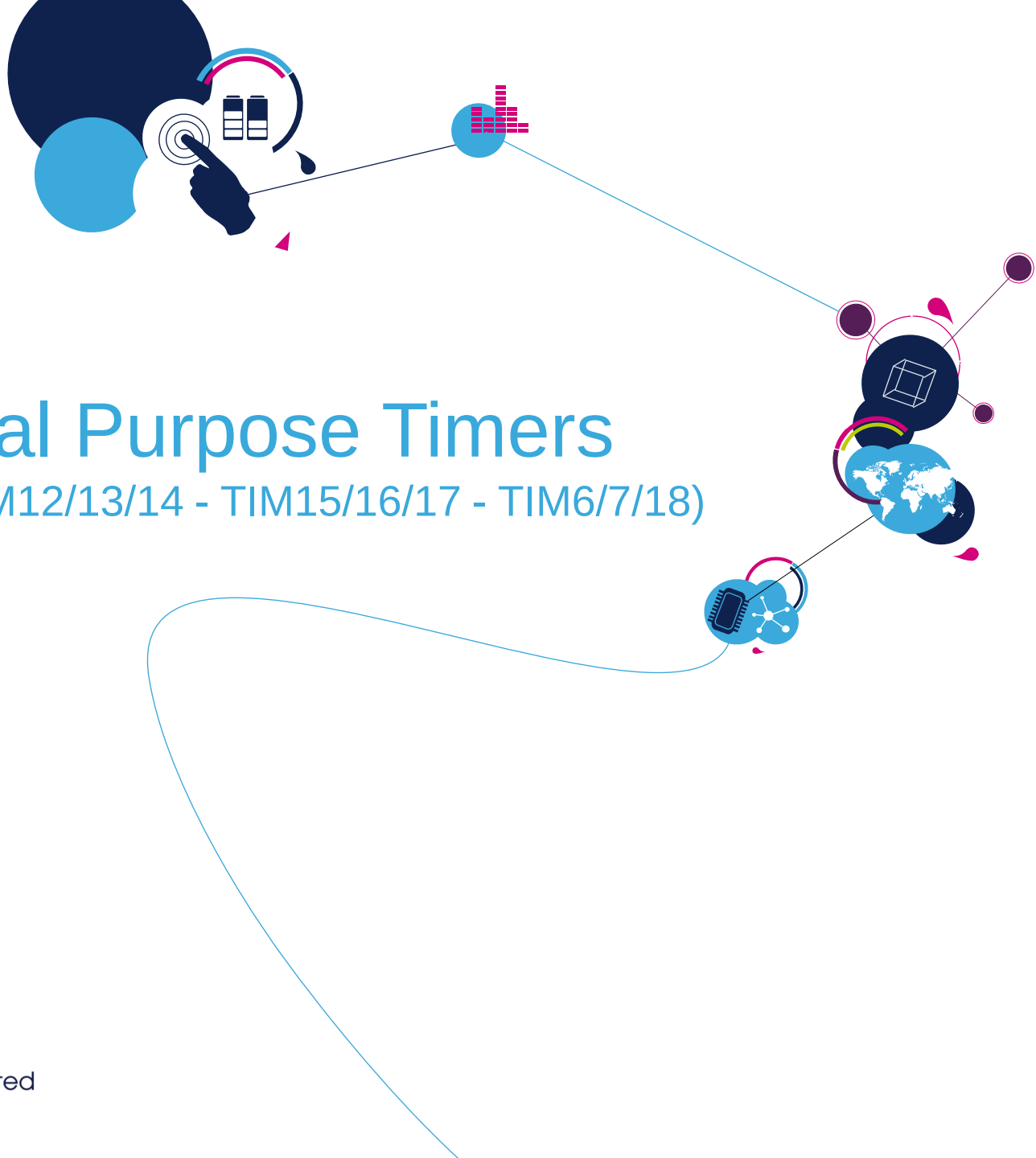


STM32F3 Technical Training

For reference only

Refer to the latest documents for details





General Purpose Timers

(TIM2/3/4/5 - TIM12/13/14 - TIM15/16/17 - TIM6/7/18)

STM32F30x Timer features overview

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	Counter resolution	Counter Type	Prescaler factor	DMA	Capture Compare channels	Synchronization	
						Master config	Slave config
General purpose TIM2	32 bit	Up, Down and Up/Down	1...65536	YES	4	YES	YES
General purpose TIM3 and TIM4	16 bit	Up, Down and Up/Down	1...65536	YES	4	YES	YES
Basic TIM6 and TIM7	16 bit	Up	1...65536	YES	0	YES	NO
1 channel, 1 complementary output TIM16 and TIM17	16 bit	Up	1...65536	NO	1	YES ⁽¹⁾	NO
2 channels, 1 complementary output TIM15	16 bit	Up	1...65536	NO	2	YES	YES

⁽¹⁾ TIM16 and TIM17 have no TRGO output, instead OC output is used

STM32F37x Timer features overview

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	Counter resolution	Counter Type	Prescaler factor	DMA	Capture Compare channels	Synchronization	
						Master config	Slave config
General purpose TIM2 and TIM5	32 bit	Up, Down and Up/Down	1...65536	YES	4	YES	YES
General purpose TIM3, TIM4 and TIM19	16 bit	Up, Down and Up/Down	1...65536	YES	4	YES	YES
Basic TIM6, TIM7 and TIM18	16 bit	Up	1...65536	YES	0	YES	NO
1 channel, 1 complementary output TIM16 and TIM17	16 bit	Up	1...65536	NO	1	YES ⁽¹⁾	NO
2 channels, 1 complementary output TIM15	16 bit	Up	1...65536	NO	2	YES	YES
1 channel TIM13 and TIM14	16 bit	Up	1...65536	NO	1	YES ⁽²⁾	NO
2 channels TIM12	16 bit	Up	1...65536	NO	2	NO	YES

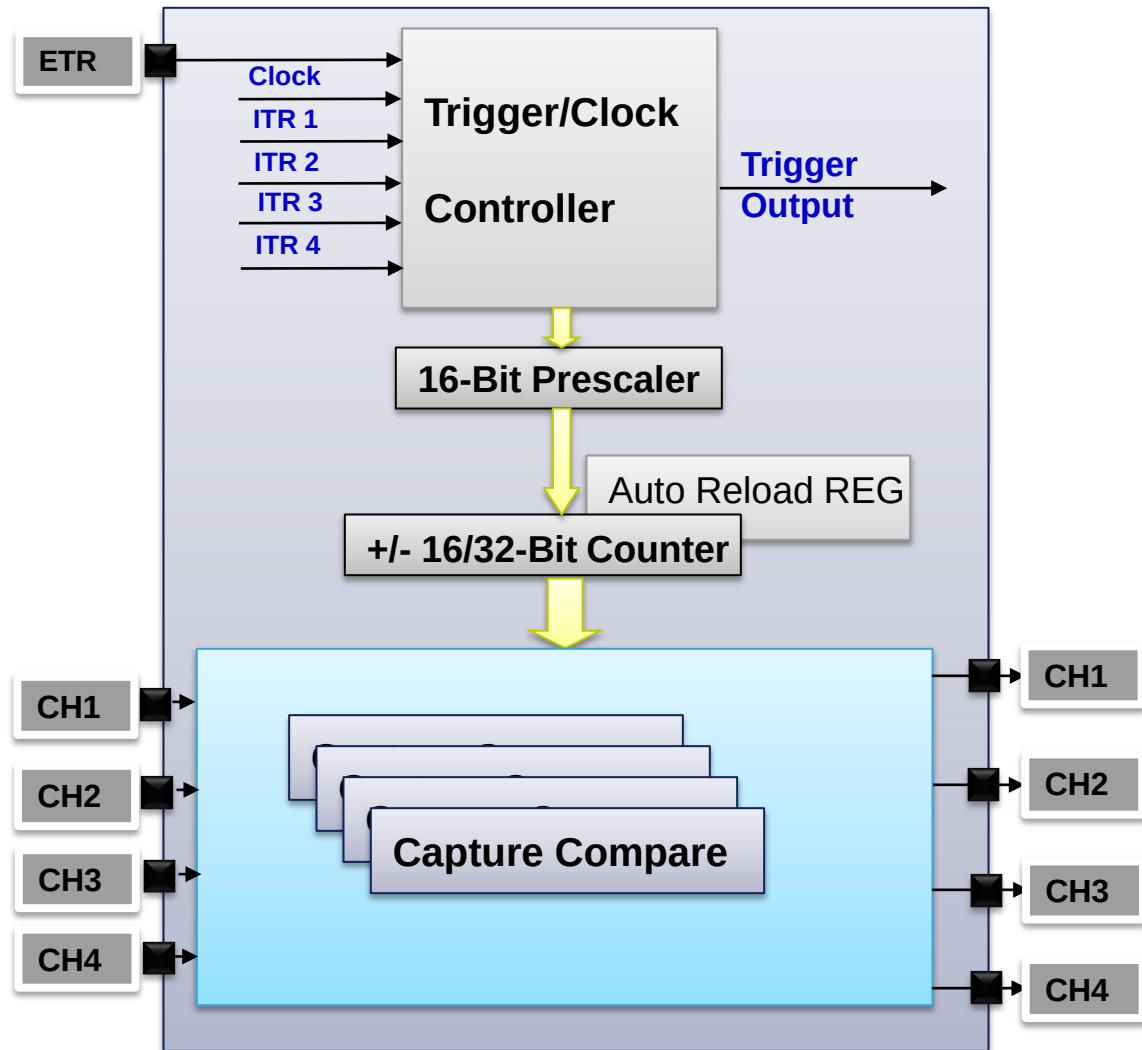
⁽¹⁾ TIM16 and TIM17 have no TRGO output, instead OC output is used

⁽²⁾ TIM13 and TIM14 have no TRGO output, instead OC output is used

Features overview (1/3)

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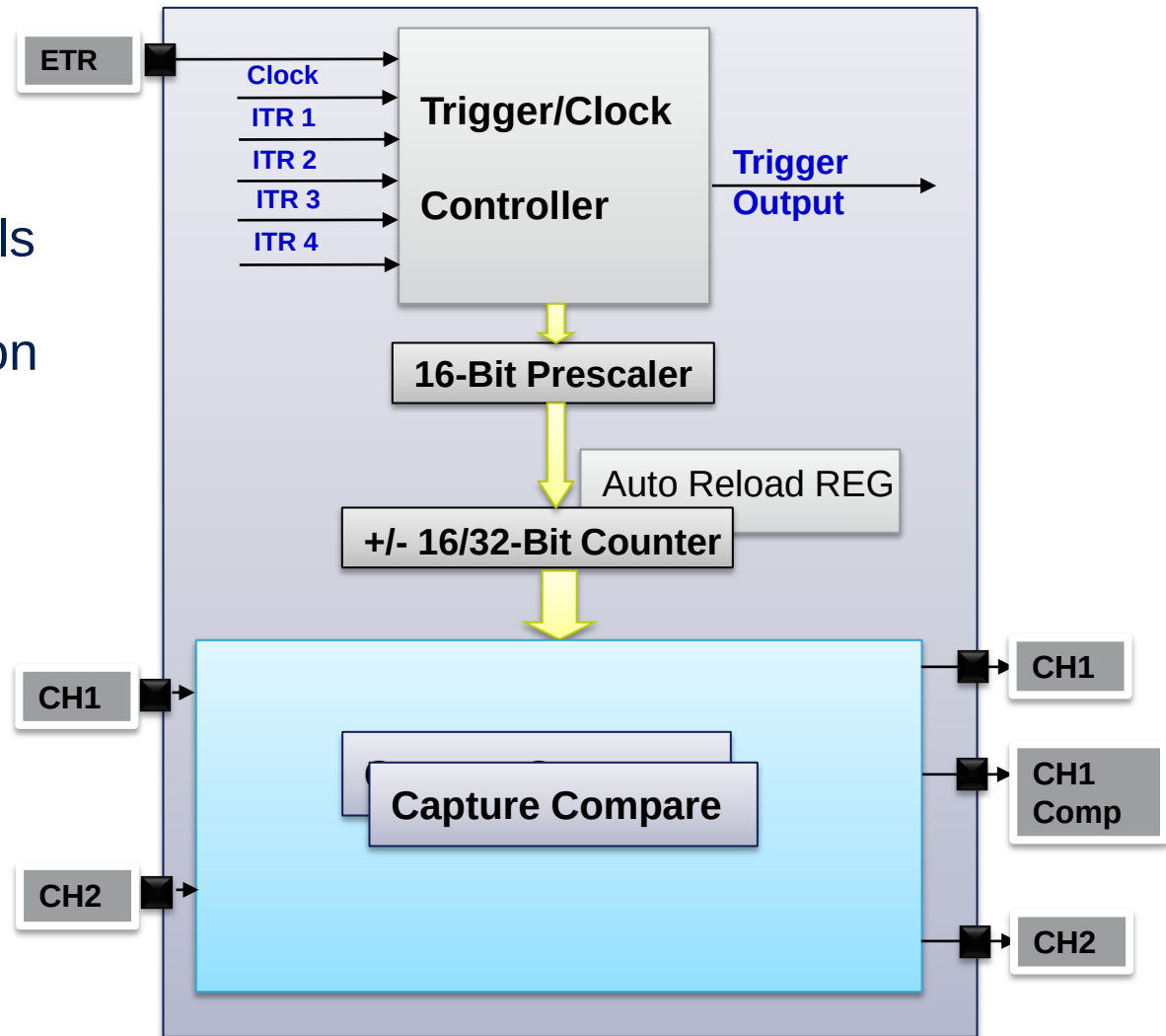
- Up to 4 16-bit resolution Capture Compare channels (TIM3/4/19)
- Up to 4 32-bit resolution Capture Compare channels (TIM2/5)
- Inter-timers synchronization
- Up to 6 IT/DMA Requests
- Encoder Interface
- Hall sensor Interface



Features overview (2/3)

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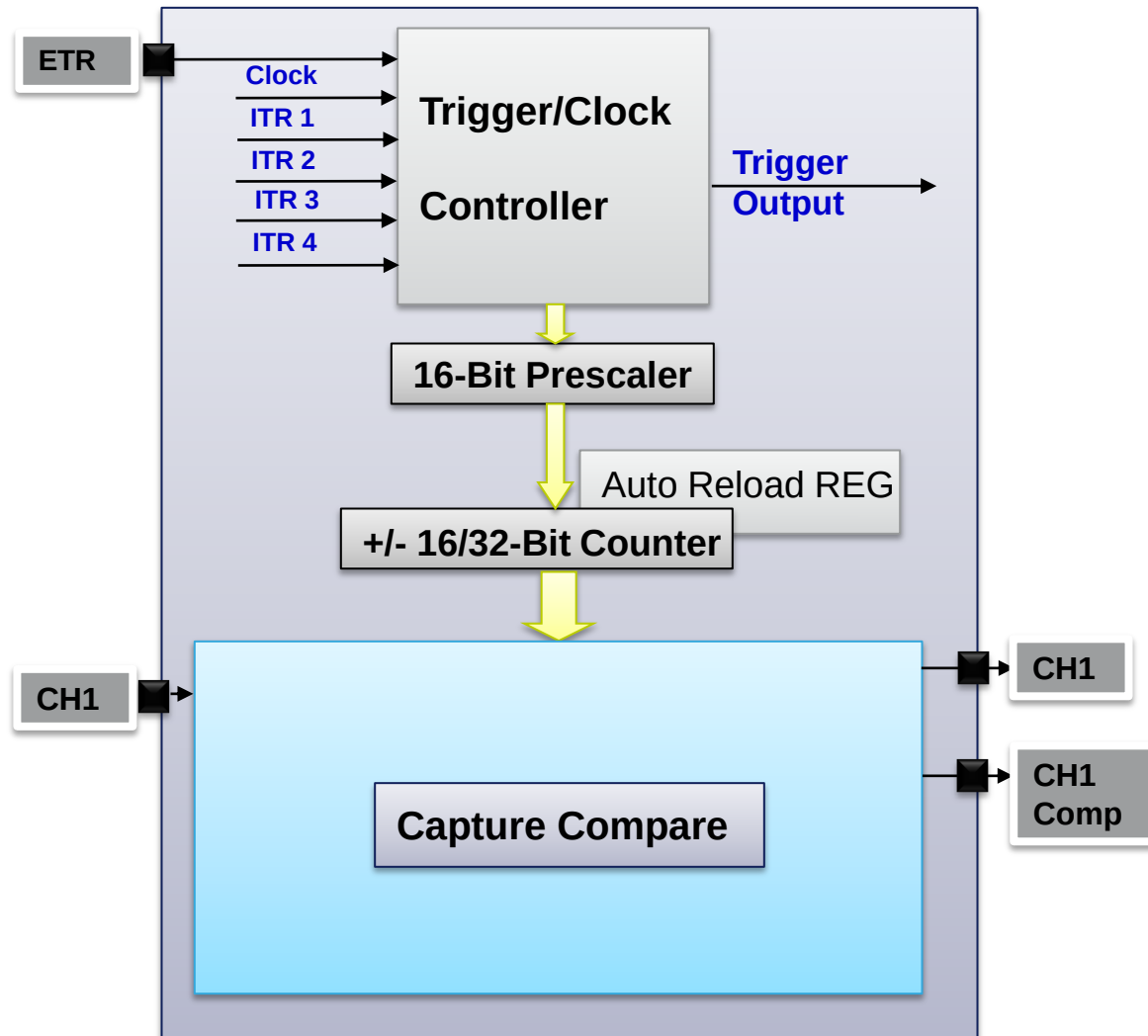
- Up to 2 16-bit resolution Capture Compare channels
- Inter-timers synchronization
- Encoder Interface
- Only TIM15 has complementary output on channel1



Features overview (3/3)

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- One 16-bit resolution Capture Compare channels
- Only TIM16/17 has complementary output on channel 1

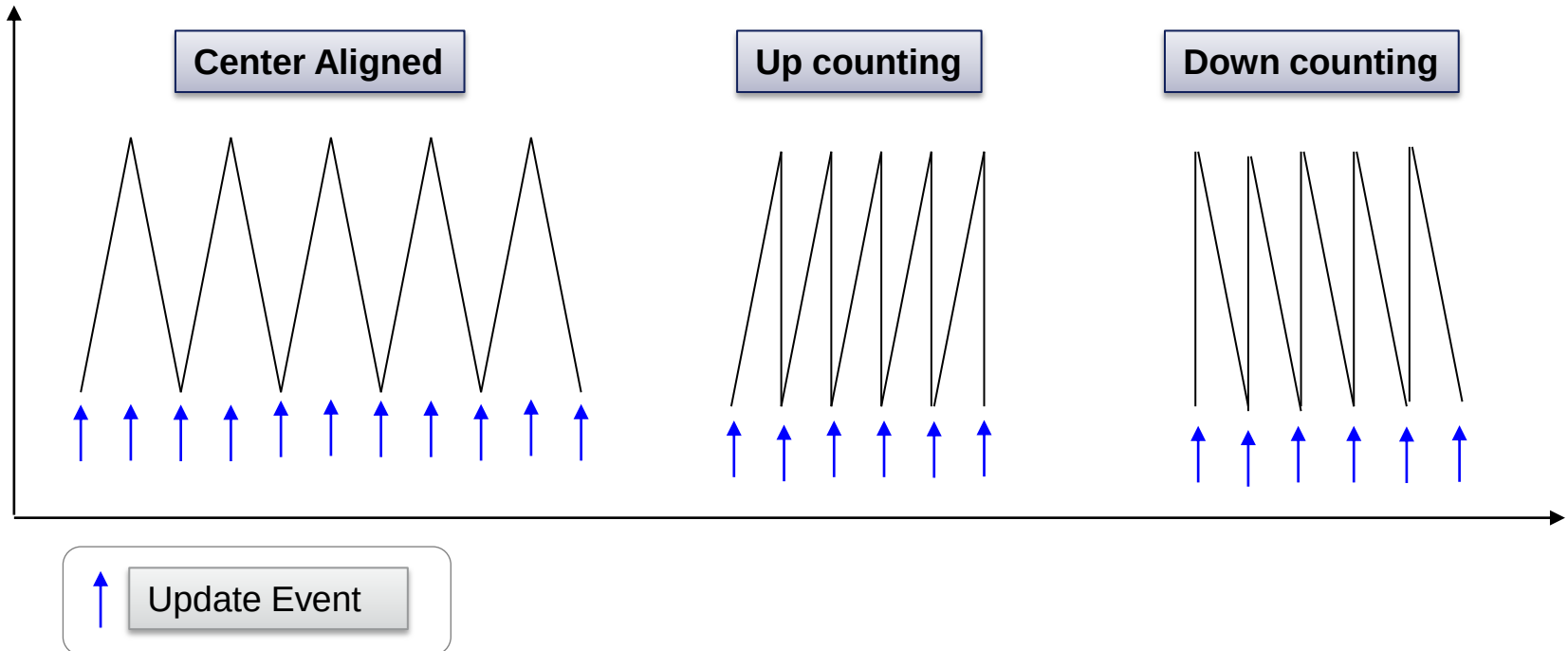


Counting Modes (1/2)

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- There are three counter modes:

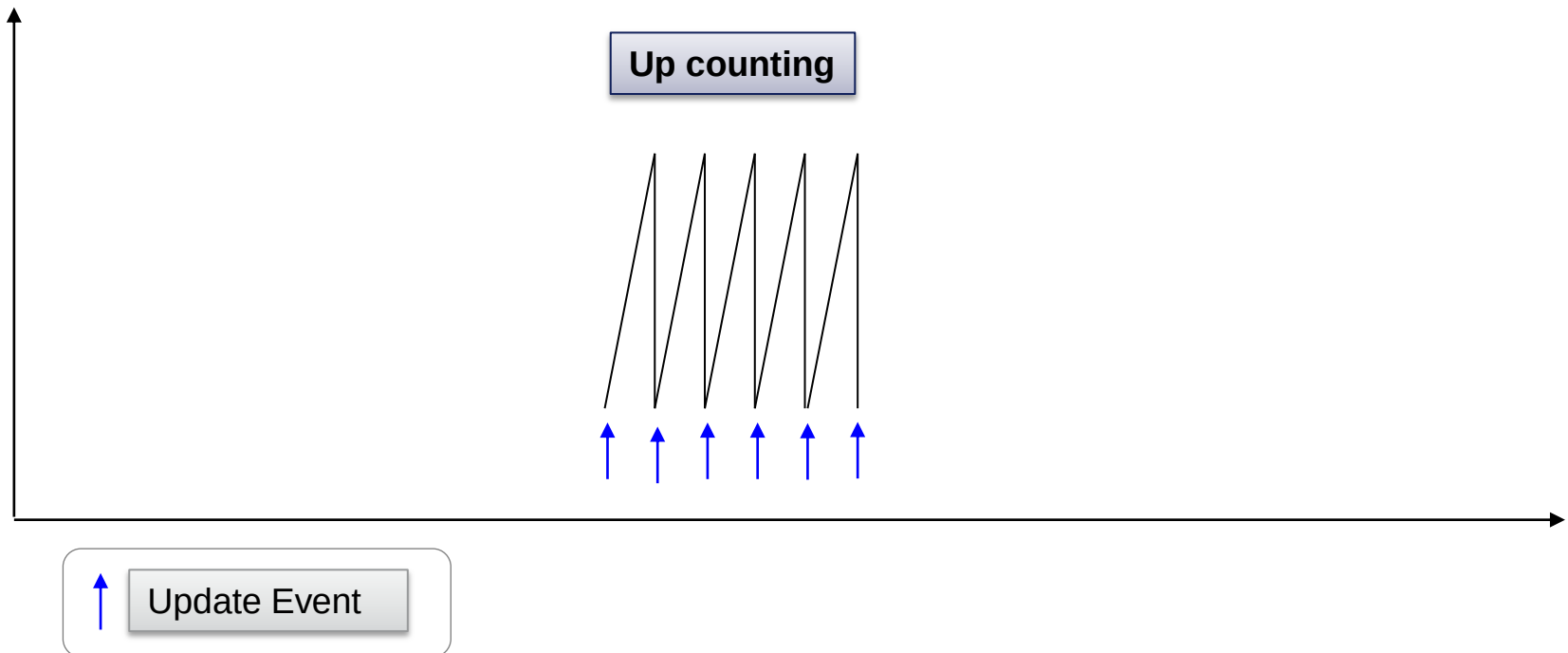
- Up counting mode
- Down counting mode
- Center-aligned mode



Counting Modes (2/2)

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- There is only one counting mode:
 - Up counting mode

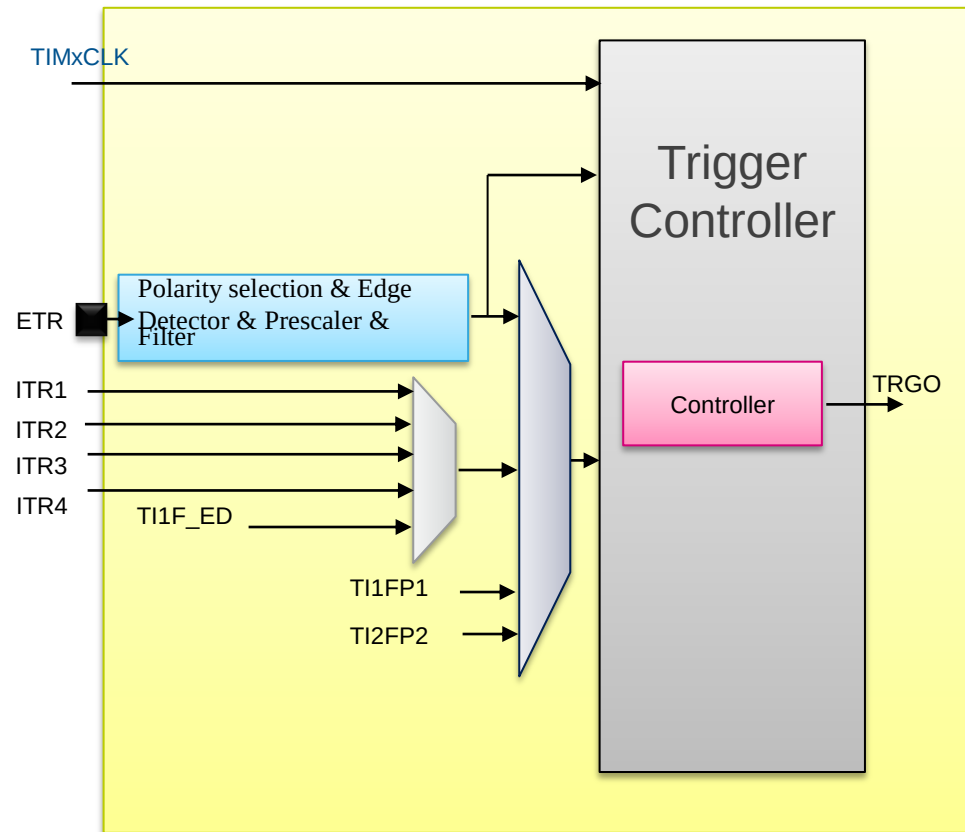


- The content of the preload register is transferred into the shadow register
 - depends on the *Auto-reload Preload* feature if enabled or not
 - If enabled, at each Update Event the transfer occurs
 - If not enabled, the transfer occurs Immediately
- The Update Event is generated
 - For each counter overflow/underflow
 - Through software, by setting the UG bit (Update Generation)
- The Update Event (UEV) request source can be configured to be
 - Next to counter overflow/underflow event
 - Nest to Counter overflow/underflow event plus the following events
 - Setting the UG bit by software
 - Trigger active edge detection (through the slave mode controller)

Counter Clock Selection

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- Clock can be selected out of 8 sources
 - Internal clock TIMxCLK provided by the RCC
 - Internal trigger input 1 to 4:
 - ITR1 / ITR2 / ITR3 / ITR4
 - Using one timer as prescaler for another timer
 - External Capture Compare pins
 - Pin 1: TI1FP1 or TI1F_ED
 - Pin 2: TI2FP2
 - External pin ETR
 - Enable/Disable bit
 - Programmable polarity
 - 4 Bits External Trigger Filter
 - External Trigger Prescaler:
 - Prescaler off
 - Division by 2
 - Division by 4
 - Division by 8



Capture Compare Array presentation

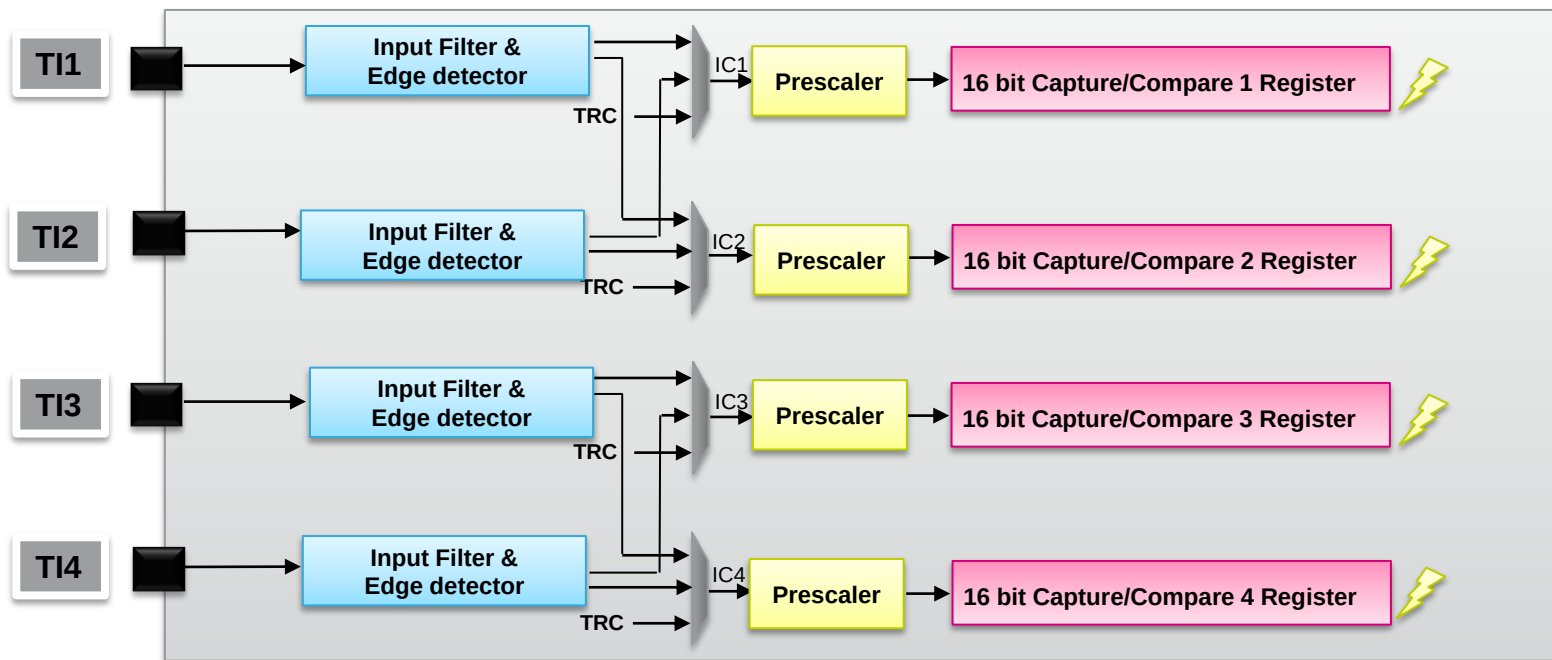
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- Up to 4 channels
 - TIM2/3/4/5/19 have 4 channels
 - TIM12/15 have 2 channels
 - TIM13/14/16/17 have one channel
 - TIM6/7/18 have no channels
- Programmable bidirectional channels
 - Input direction: channel configured in Capture mode
 - Output direction: Channel configured in Compare mode
- Channel's main functional blocs
 - Capture/Compare register
 - Input stage for capture
 - 4-bit digital filter
 - Input Capture Prescaler:
 - Output stage for Compare
 - Output control bloc

Input Capture Mode (1/2)

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- Capture stage architecture



Input Capture Mode (2/2)

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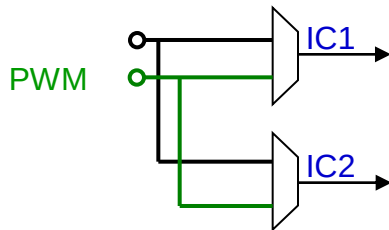
- Flexible mapping of TIx inputs to channels' inputs ICx
 - {TI1->IC1}, {TI1->IC2}, {TI2->IC1} and {TI2->IC2} are possible
- When an active Edge is detected on ICx input, the counter value is latched in the corresponding CCR register.
- When a Capture Event occurs, the corresponding CCXIF flag is set and an interrupt or a DMA request can be sent if they are enabled.
- An over-capture flag for over-capture signaling
 - Takes place when a Capture Event occurs while the CCxIF flag was already high

PWM Input Mode

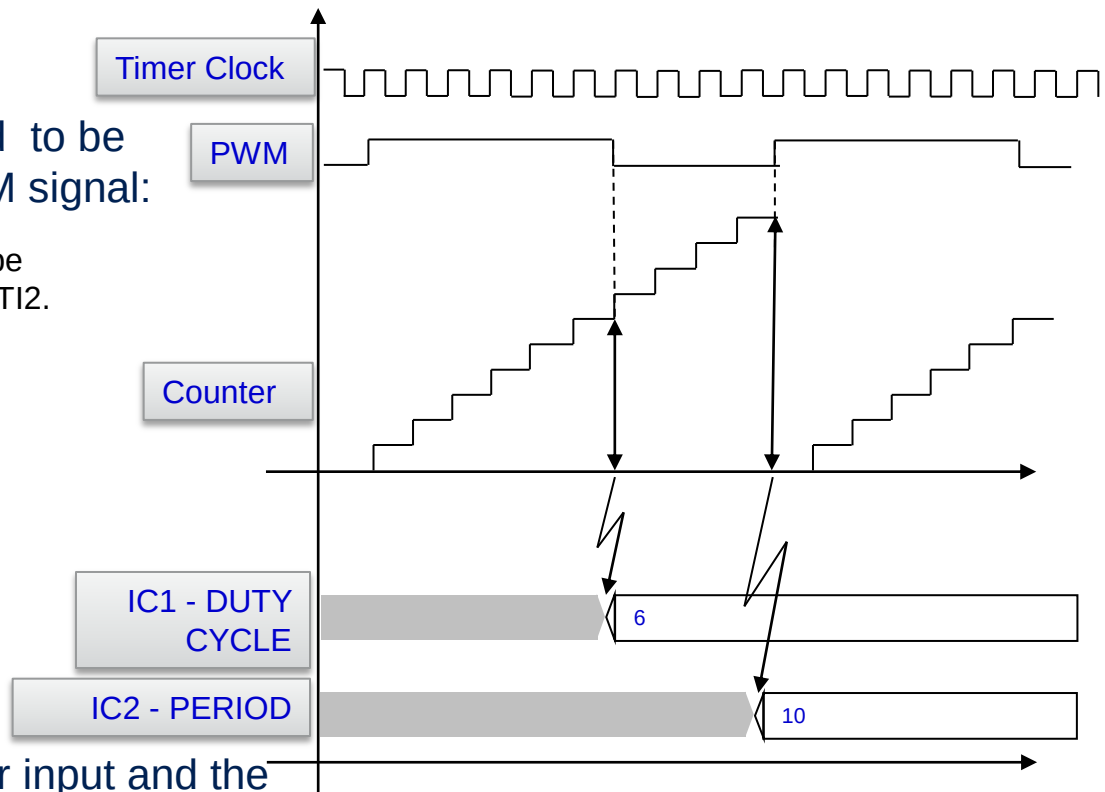
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- IC1 and IC2 must be configured to be connected together to the PWM signal:

⇒ IC1 and IC2 are redirected internally to be mapped to the same external pin TI1 or TI2.



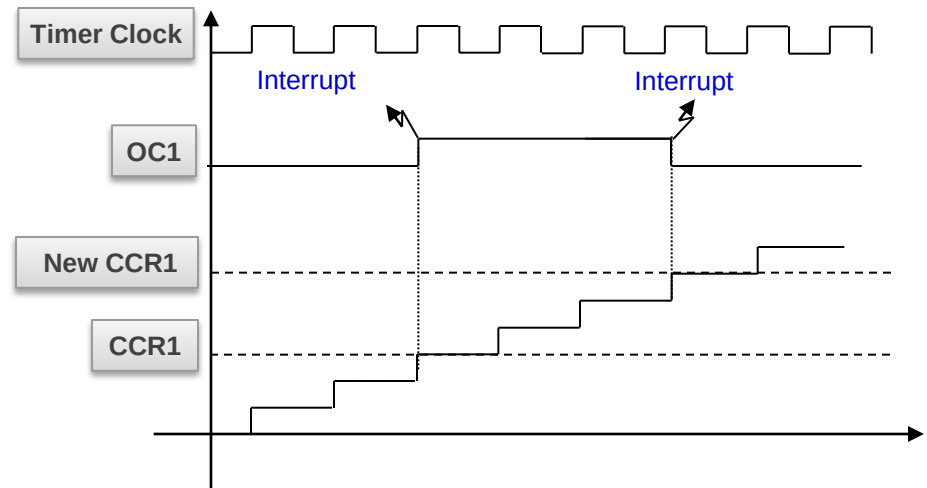
- IC1 and IC2 active edges must have opposite polarity.
- IC1 or IC2 is selected as trigger input and the slave mode controller is configured in reset mode.
- The PWM Input functionality enables the measurement of the period and the pulse width of an external waveform.



Output Compare Mode

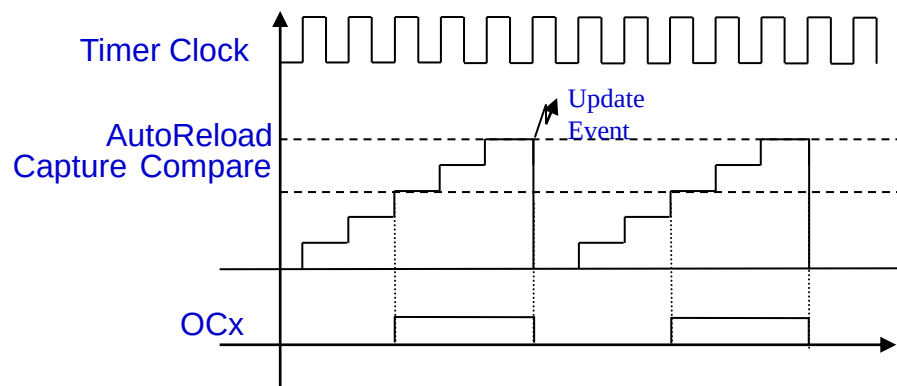
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- The Output Compare is used to control an output waveform or indicate when a period of time has elapsed.
- When a match is found between the capture/compare register and the counter:
 - The corresponding output pin is assigned to the programmable Mode, it can be:
 - Set
 - Reset
 - Toggle
 - Remain unchanged
 - Set a flag in the interrupt status register
 - Generates an interrupt if the corresponding interrupt mask is set
 - Send a DMA request if the corresponding enable bit is set
- The CCRx registers can be programmed with or without preload registers

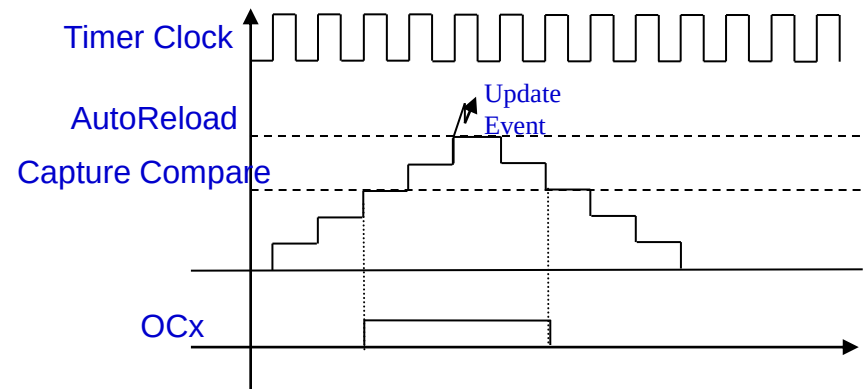


- Available on all channels
- Two PWM mode available
 - PWM mode 1
 - PWM mode 2
 - Each PWM mode behavior (waveform shape) depends on the counting direction

Edge-aligned Mode



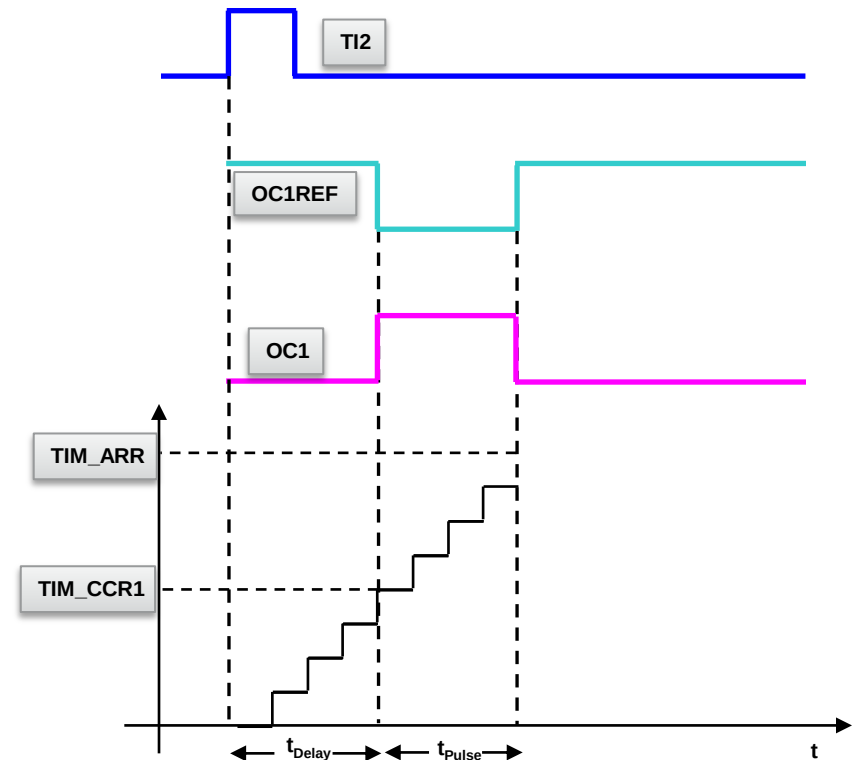
Center-aligned Mode



One Pulse Mode (1/2)

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- One Pulse Mode (OPM) is a particular case of Output Compare mode
- It allows the counter to be started in response to a stimulus and to generate a pulse
 - With a programmable length
 - After a programmable delay
- There are two One Pulse Mode waveforms selectable by software:
 - Single Pulse
 - Repetitive Pulse



One Pulse Mode (2/2)

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Exercise:

How to configure One Pulse Mode to generate a repetitive Pulse in response to a stimulus ?

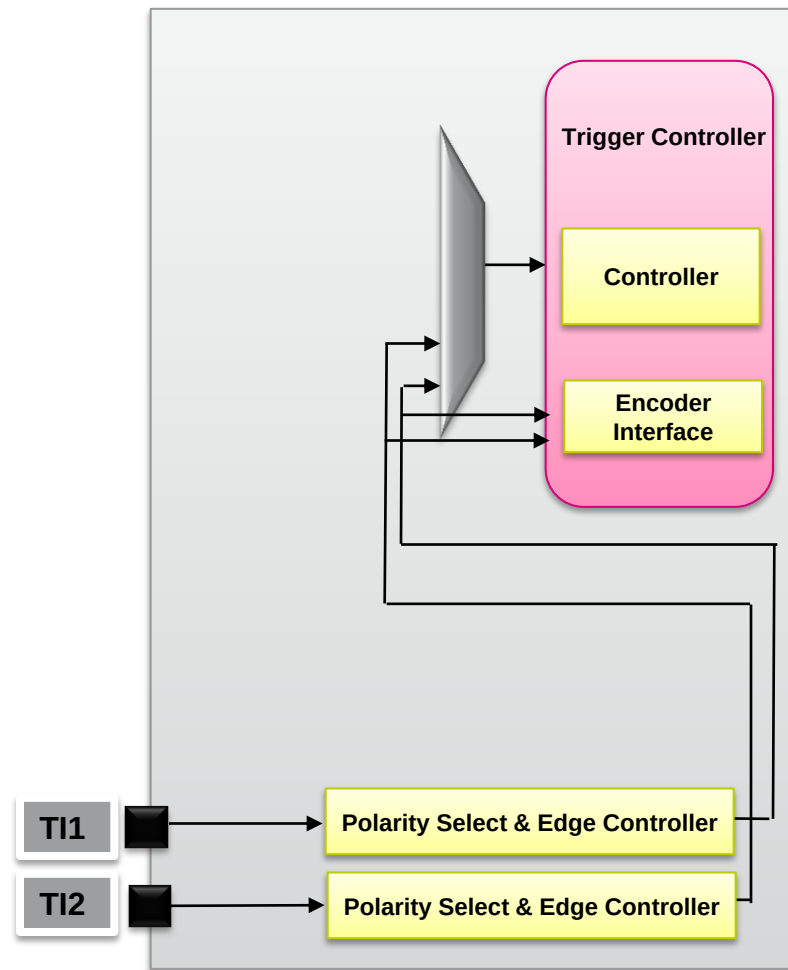
One Pulse Mode configuration steps

1. **Input Capture Module Configuration:**
 - i. Map TlxFPx on the corresponding TlX.
 - ii. TlxFPx Polarity configuration.
 - iii. TlxFPx Configuration as trigger input.
 - iv. TlxFPx configuration to start the counter (Trigger mode)
2. **Output Compare Module Configuration:**
 - i. OCx configuration to generate the corresponding waveform.
 - ii. OCx Polarity configuration.
 - iii. t_{Delay} and t_{Pulse} definition.
3. **One Pulse Module Selection:** Set or Reset the corresponding bit (OPM) in the Configuration register (CR1).

Encoder Interface (1/2)

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- Encoders are used to measure position and speed of mobile systems (either linear or angular)
- The encoder interface mode acts as an external clock with direction selection
- Encoders and Microcontroller connection example:
 - A can be connected directly to the MCU without external interface logic.
 - The third encoder output which indicates the mechanical zero position, may be connected to an external interrupt and trigger a counter reset.
- Encoder enhancement
 - A copy of the Update Interrupt Flag (UIF) is copied into bit 31 of the counter register
 - Simultaneous read of the Counter value and the UIF flag : Simplify the position determination



Encoder Interface (2/2)

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Exercise:

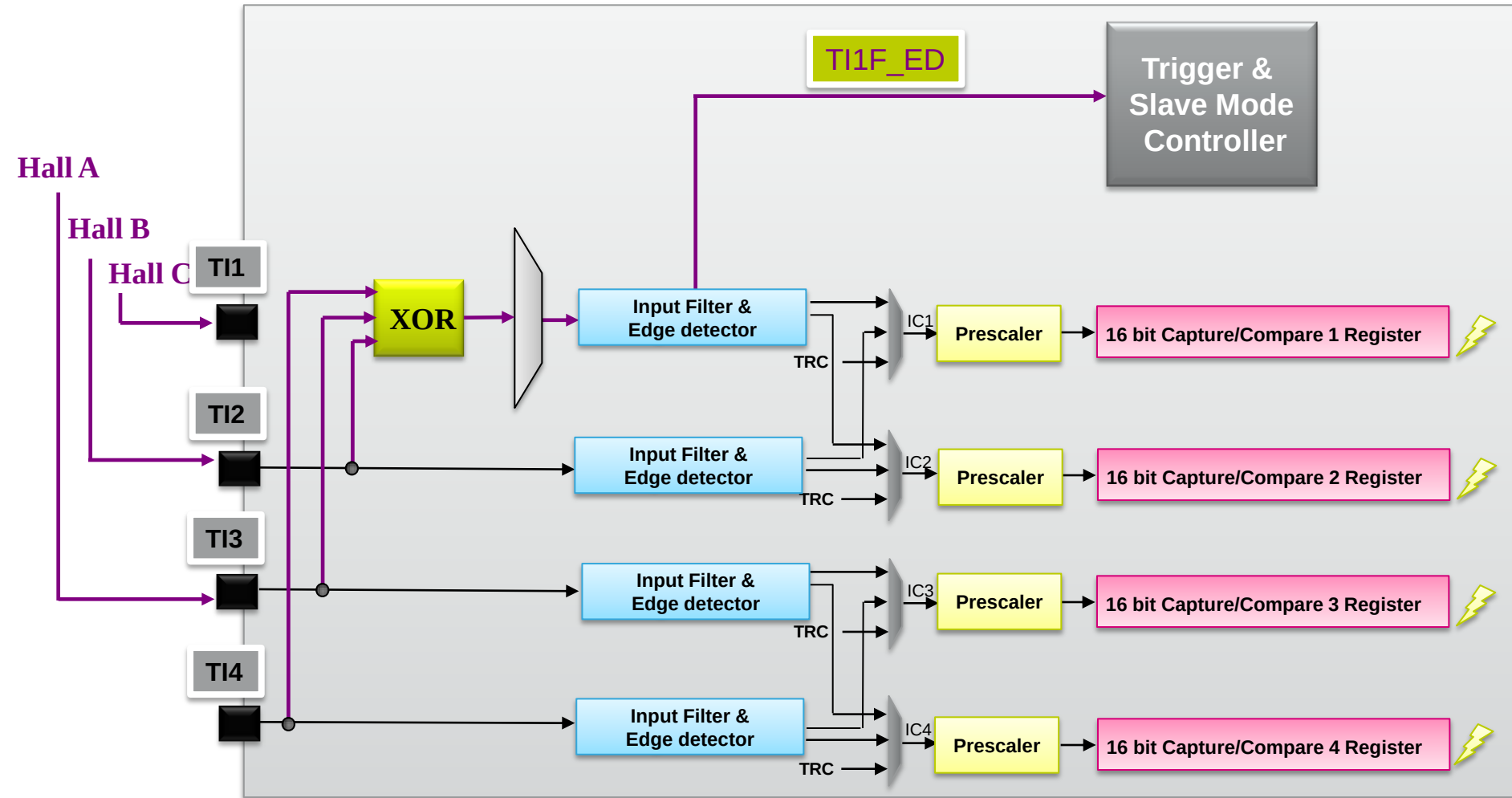
How to configure the Encoder interface to detect the rotation direction of a motion system?

Encoder interface configuration steps:

1. Select the active edges: example counting on TI1 and TI2.
2. Select the polarity of each input: example TI1 and TI2 polarity not inverted.
3. Select the corresponding Encoder Mode.
4. Enable the counter.

Hall sensor Interface (1/2)

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Hall sensor Interface (2/2)

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- Hall sensors are used for:
 - Speed detection
 - Position sensor
 - Brushless DC Motor Sensor
- How to configure the TIM to interface with a Hall sensor?
 - Select the hall inputs for TI1: TI1S bit in the CR2 register
 - The slave mode controller is configured in reset mode
 - TI1F_ED is used as input trigger
- To measure a motor speed:
 - Use the Capture/Compare Channel 1 in Input Capture Mode
 - The Capture Signal is the TRC signal
 - The captured value which correspond to the time elapsed between 2 changes on the inputs, gives an information about the motor speed

Synchronization Mode Configuration

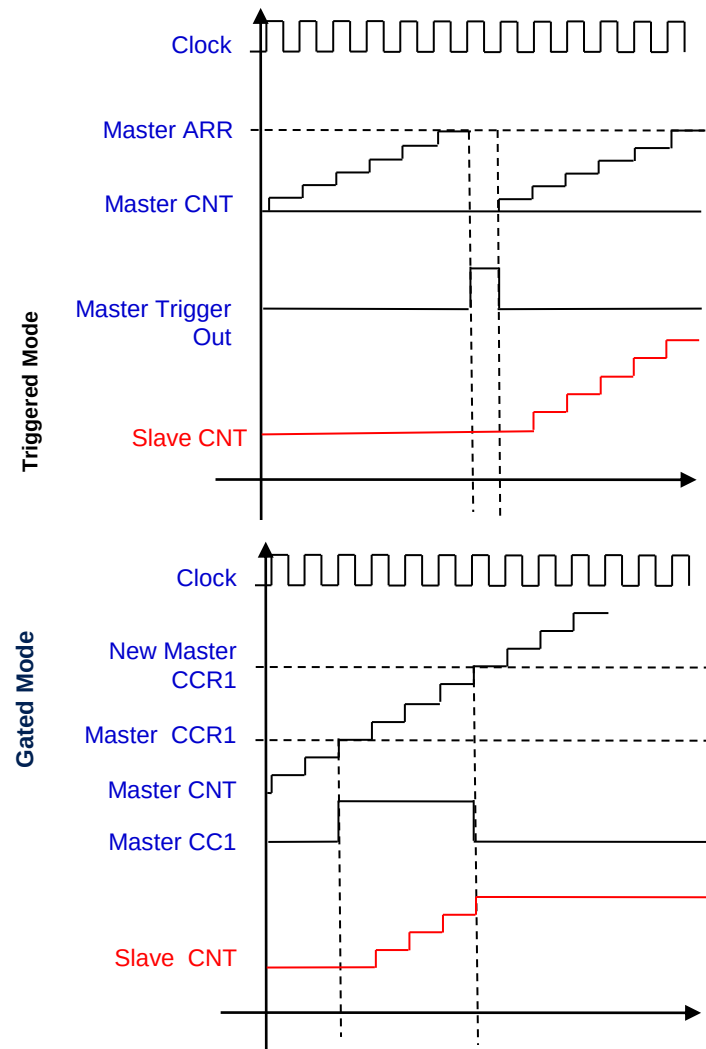
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- The Trigger Output can be controlled on:

- Counter reset
- Counter enable
- Update event
- OC1 / OC1Ref / OC2Ref / OC3Ref / OC4Ref signals

- The slave timer can be controlled in two modes:

- Triggered mode : only the start of the counter is controlled
- Gated Mode: Both start and stop of the counter are controlled
- Reset Mode - Rising edge of the selected trigger input (TRGI) reinitializes the counter

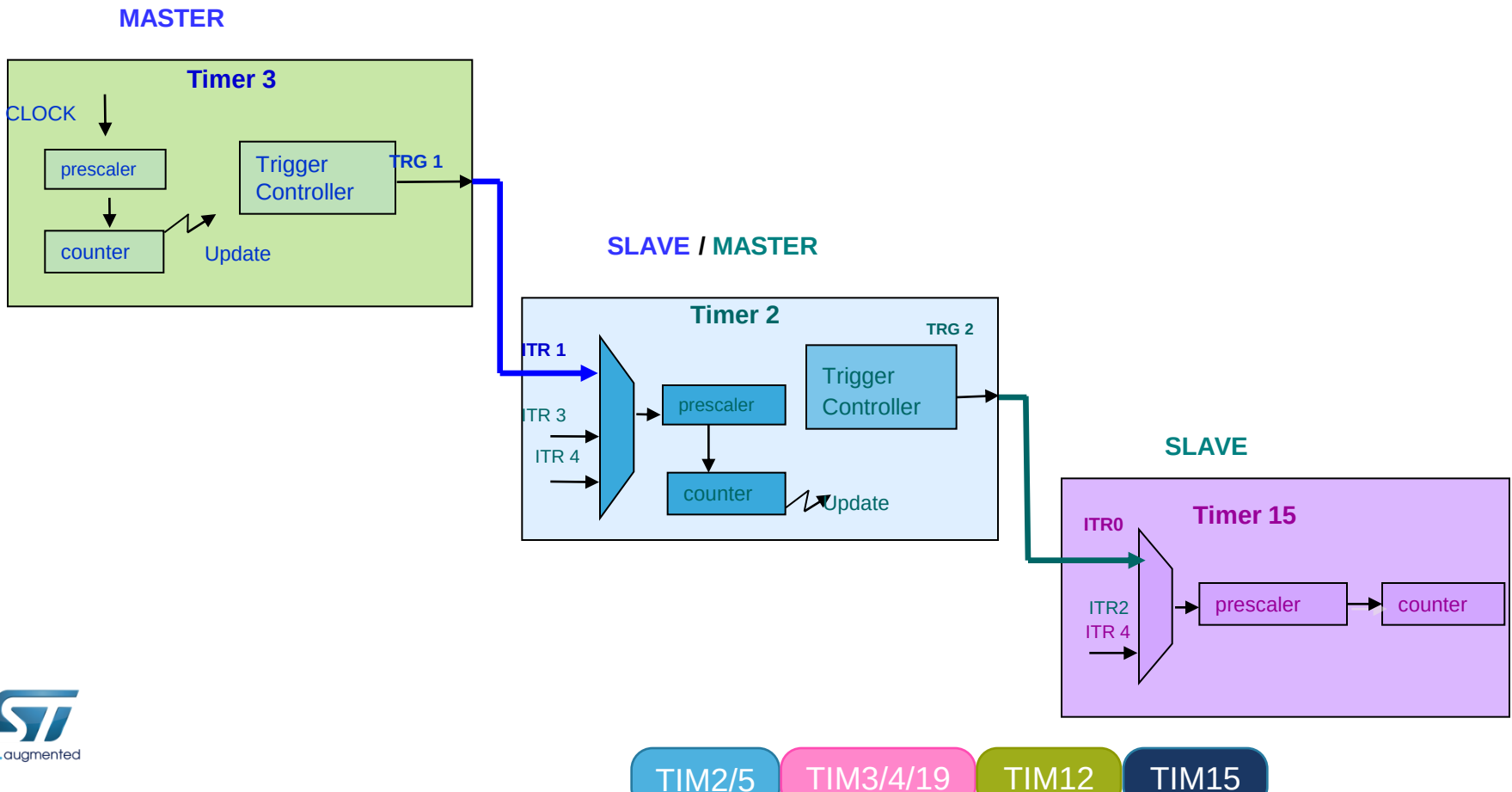


Synchronization: Configuration examples (1/3)

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- Cascade mode:

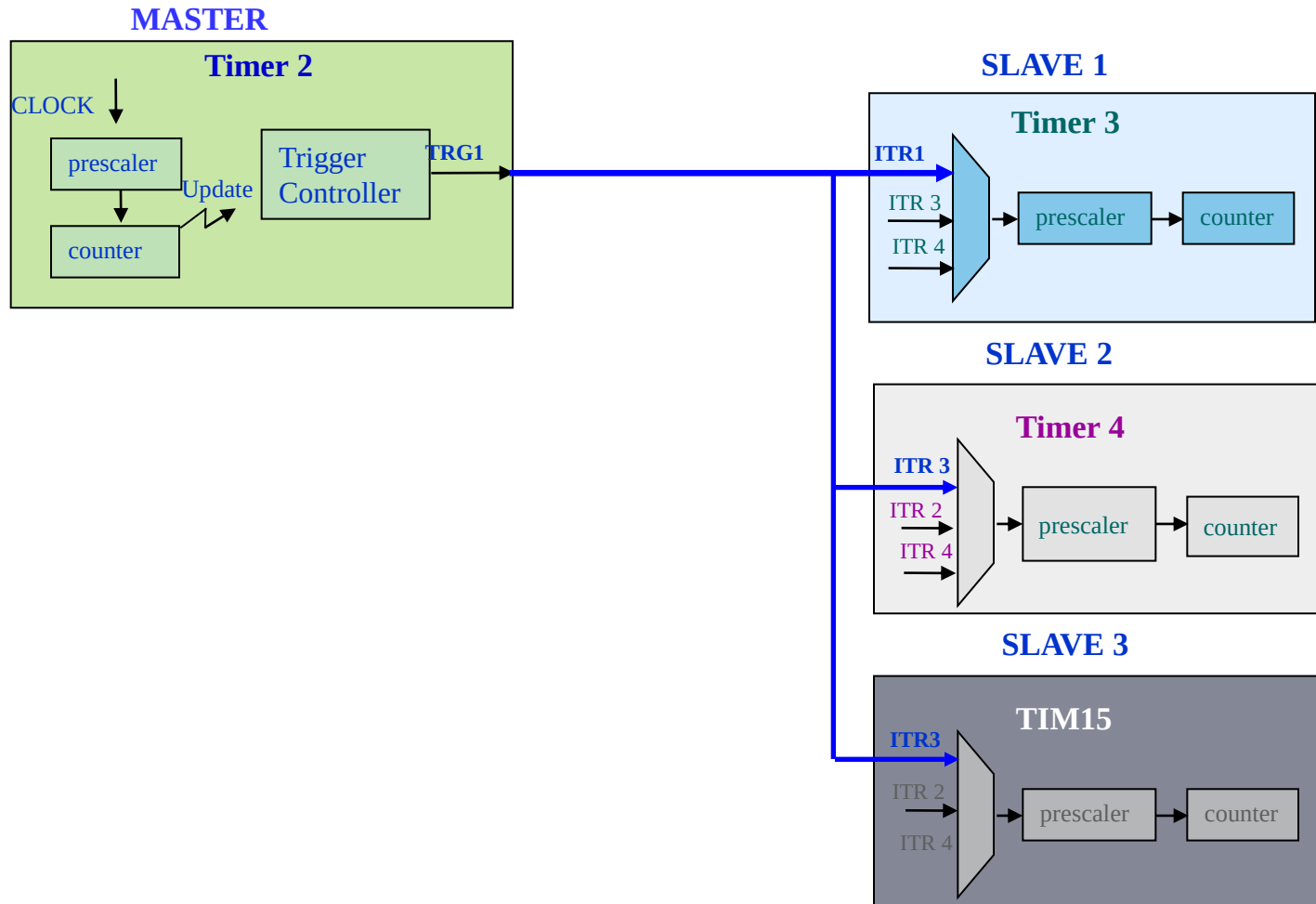
- TIM3 used as master timer for TIM2
- TIM2 configured as TIM3 slave, and master for TIM15



Synchronization: Configuration examples (2/3)

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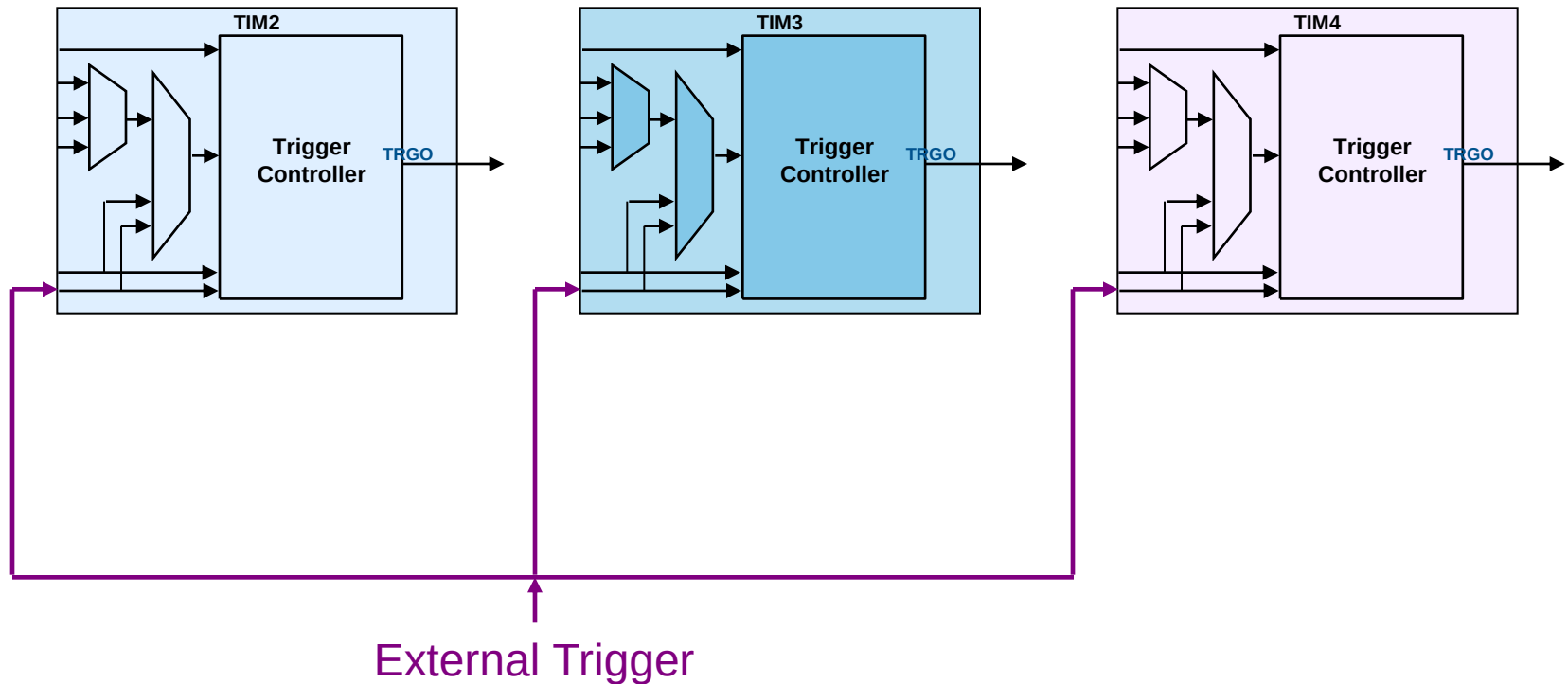
- One Master several slaves: TIM2 used as master for TIM3, TIM4 and TIM15



Synchronization: Configuration examples (3/3)

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- Timers and external trigger synchronization
 - TIM2, TIM3 and TIM4 are slaves for an external signal connected to respective Timers inputs





Thank you !

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